KITBASH

# User Guide

Assemble AI agents from reusable parts, prove they work, and put them in your projects.

Written against the repository as of 9 November 2026

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# 1. What Kitbash does

Kitbash builds AI agents out of small, reusable parts and tests them before they join a project. You describe an agent as a list of parts. Kitbash checks that the parts fit, runs the agent against test tasks, and writes the result into your project as Claude Code subagents or plugins.

- **Reuse.** One skill or behaviour rule serves many agents, so you improve it in one place.
- **Early errors.** The validator rejects a bad combination before any model is called, so mistakes cost nothing.
- **Evidence.** Every agent carries test tasks called evals, and a second model grades the results. A release refuses any agent without a current pass.
- **Least privilege.** Tools attach to a part of your project, not to an agent. An agent working on firmware never sees the tools for your design files.
- **Easy swaps.** Agents ask for a capability such as `source-control`, never a vendor. Changing the tool behind it is a one-line edit.

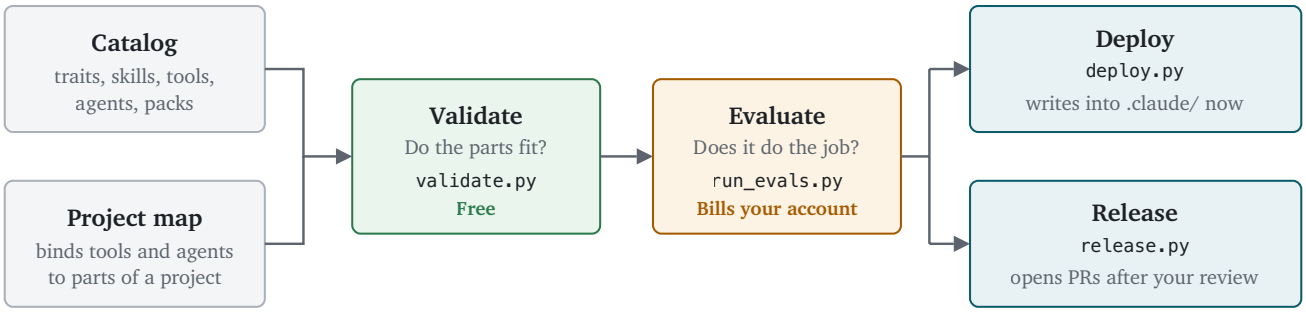



Figure 1. The path from parts to a working agent in your project.

## How to use this guide

New to Kitbash? Read chapters 2 to 4 and run the commands as you go, then chapter 5 to see a whole project put together. Part II is a set of independent tasks, so go straight to the one you need. Part III is for looking things up. Commands are marked by cost:

**Free: no model call**

```
/usr/bin/python3 scripts/validate.py
```

**Bills your account through the claude CLI**

```
/usr/bin/python3 scripts/run_evals.py security-reviewer
```

## What exists today

| Available now | Not built yet |
|---|---|
| Catalog, validator, eval runner | Automatic project maps. You write a map, or the `/kitbash` skill drafts one with Claude for you to confirm. |
| Meta-agent that designs a new agent from a sentence | A `csv-index` registry source for listing datasets. |
| Playbook runner, registry of imported parts | MCP tools in plugins. `release.py` refuses a map that binds one, so use `deploy.py` for those maps. |
| Direct deploy, reviewed release, and a Claude Code plugin with a `/kitbash` skill | |

> **A deploy belongs to one machine**
> Tool files name the programs they run through environment variables with macOS defaults, such as `KICAD_CLI`, so set those on another system (chapter 18). Deployed agent files inline the resolved paths, so a deploy is valid only on the machine that ran it.

# 2. Set up

There are two ways in. Install Kitbash as a Claude Code plugin to give one project its agents, with Claude doing the steps and asking you at each decision. Clone the repo to run the scripts yourself or to change the catalog. The rest of this guide works from a clone.

## What you need

| Requirement | Needed for |
|---|---|
| Claude Code (`claude`), signed in | The plugin, evals, playbooks and `make_agent.py` |
| Python 3.9 or later | Every script. Kitbash bundles PyYAML, so there is nothing to install. |
| Git, with network access | The first fetch of a pack or import source, and `registry.py sync` |
| GitHub CLI (`gh`), signed in | `release.py publish`, and agents that use the `github` tool |
| Domain tools such as KiCad or an STM32 toolchain | Only the agents that use them. See chapter 14. |

## Install the plugin

> **Free: in a terminal, with Claude Code 2.1.275 or later**

```
claude plugin install kitbash --marketplace manasXP/kitbash
```

Inside a session, type the same command as `/plugin install kitbash --marketplace manasXP/kitbash`. On an older Claude Code, add the marketplace first:

> **Free: older Claude Code**

```
claude plugin marketplace add manasXP/kitbash
claude plugin install kitbash@kitbash
```

The plugin carries the whole catalog and its scripts. Claude Code replaces that copy on every update, so keep nothing of your own in it.

## Set up a project with the plugin

Start Claude Code at the project's root folder and run the skill:

> **In Claude Code**

```
/kitbash:kitbash set up agents for this repo
```

1. **It looks.** Claude inventories the project and reads the catalog's agents, tools and packs.
2. **It asks.** It proposes aspects, then asks in one dialog which pack to attach, whether the project spans other repositories, and whether the aspects are right. It writes nothing before you answer.
3. **It writes and validates.** It writes `.kitbash/project-map.yaml` and runs the validator until it reports `0 error(s)`.
4. **It evaluates, on your go-ahead.** For each routed agent it shows the dry run, checks that the tools the evals need are installed, and asks before each billed run, naming its ceiling.

5. **It deploys what passed.** It runs the deploy dry run, and deploys only once every routed agent has a current pass.

The skill pre-approves its free commands, so the validator and the dry runs run without a permission prompt. Writing files and every billed command still ask. Everything that belongs to the project goes in its `.kitbash/` folder:

**A project's .kitbash/ folder**

```
.kitbash/
  project-map.yaml     aspects, the tools bound to each, the agents routed to each
  playbooks/           optional multi-agent sequences
  catalog/             optional project agents, traits, skills, tools, hooks, fixtures
  evals.lock.json      eval results, written by run_evals.py and read by deploy.py
  work/                eval workspaces, ignored by git
```

Commit `.kitbash/` and the files the deploy writes in `.claude/`. A map in `.kitbash/` needs no `catalog:` line, since it uses the catalog of the scripts that read it. Project agents in `.kitbash/catalog/` must not reuse a catalog name.

## Run the scripts from a clone

**Free: run once**

```
git clone https://github.com/manasXP/kitbash.git ~/kitbash
cd ~/kitbash/catalog
```

Commands in the rest of this guide run from `catalog/`. The same scripts also work inside a project with a `.kitbash/` folder, from its root or any folder below it. There they find the map, merge `.kitbash/catalog/` into the catalog, and need no `--map`:

**Free: run inside the project**

```
/usr/bin/python3 ~/kitbash/catalog/scripts/validate.py
/usr/bin/python3 ~/kitbash/catalog/scripts/run_evals.py security-reviewer --aspect api --dry-run
/usr/bin/python3 ~/kitbash/catalog/scripts/deploy.py --dry-run
```

**Which Python to use**

This guide writes `/usr/bin/python3`, the macOS system interpreter. Any Python 3.9 or later works, so substitute its path on other systems. Tool commands run under the same interpreter as Kitbash. The `recipe-validator` tool also needs pydantic, so use an interpreter that has it, or point `KITBASH_PYTHON` at one.

## Check that it works

**Free**

```
/usr/bin/python3 scripts/validate.py
```

The first run fetches the pinned skill packs. The last line of a clean first run looks like this:

**Output on a fresh clone**

```
0 error(s), 4 warning(s); 11 agent(s), 1 project map(s), 2 playbook(s) checked
```

Three of the warnings are permanent and harmless: the `gsd`, `gstack` and `super-board` packs have empty bases on purpose. The fourth says the registry isn't synced. Clear it now with:

**Free: uses the network, no model. Takes under two minutes.**

```
/usr/bin/python3 scripts/registry.py sync
/usr/bin/python3 scripts/validate.py
```

> **Output once synced: this is the clean state**
>
> `0 error(s), 3 warning(s); 11 agent(s), 1 project map(s), 2 playbook(s) checked`

### What costs money

| Cost | Commands | Limit |
|---|---|---|
| Free | `validate.py`, `registry.py`, `deploy.py`, every `--dry-run` except the one below, and `release.py build`, `check` and `publish` | None needed |
| Bills | `run_evals.py` and `run_playbook.py` | `--budget` per agent run or playbook step, default $1.00. The judge gets $0.20 per eval. |
| Bills | `make_agent.py`, **including with `--dry-run`**, which still makes the design call | `--budget` for the design call, default $1.00, plus any evals it runs |
| Bills | `release.py evals` | `--max-total`, default $30. It stops starting new agents at this ceiling. |

The caps let you work out a ceiling before you run anything. An agent with five evals can spend at most 5 × ($1.00 + $0.20) = $6.00 at the defaults. Recorded runs have come in well under the caps: the two sample playbooks each passed both of their evals, for $0.22 and $0.54.

### Where Kitbash keeps things

| Location | Contents |
|---|---|
| `catalog/` | Parts and fixtures, with the scripts in `catalog/scripts/` |
| `.kitbash/` in your project | The project's map, playbooks and own parts, its eval results in `evals.lock.json`, and eval workspaces in `work/` |
| `projects/<name>/` | Project maps and playbooks you can publish |
| `private/projects/<name>/` | A separate private repo for project content. Git ignores this folder. |
| `~/.cache/kitbash/packs/`, `sources/` | Pinned checkouts of packs and imported skills |
| `~/.local/share/kitbash/registry.db` | The SQLite registry of imported skills and MCP servers |

Everything under your home folder is disposable. Delete it and Kitbash rebuilds it from the repo's files.

# 3. A quick tour

Steps 1 to 5 are free and change nothing. They show you what Kitbash knows and what an agent is actually made of. Run them from `catalog/`.

#### Step 1. See what an agent is made of

`--dry-run` prints the exact prompt an agent receives, then the commands that would run for each eval, without calling a model.

> **Free**
>
> `/usr/bin/python3 scripts/run_evals.py security-reviewer --dry-run`

**Output, shortened**

```
===== security-reviewer: composed system prompt =====
## Trait: read-only
You may inspect the codebase but never modify it. Report findings; do not fix them.
...
## Trait: concise-reporter
Lead with the conclusion. One sentence per finding. Reference code as path:line.
...
## Skill: security-review
# Security review
...
## Agent: security-reviewer
You review pull requests for security defects only. Ignore style, naming, ...
----- denied tools: ['Edit', 'Write']  hooks: ['.../block-network.py']  mcp: []  pack: None
```

The prompt is simply the parts joined in order. The agent can't edit files because its `read-only` trait denies `Edit` and `Write`.

### Step 2. See what a project adds

Add a map and an aspect, and the agent is composed the way that project would run it.

**Free**

```
/usr/bin/python3 scripts/run_evals.py security-reviewer --dry-run \
  --map ../projects/smart-thermostat/project-map.yaml --aspect api
```

The prompt now starts with `## Pack superpowers, skill: verification-before-completion`, because that map attaches the `superpowers` pack and a pack's text comes first.

### Step 3. Search for parts you could import

**Free**

```
/usr/bin/python3 scripts/registry.py search "pdf forms" --limit 2
/usr/bin/python3 scripts/registry.py show github-mcp-server
```

**Output of the search, shortened**

```
skill  pdf                          anthropics-skills  Use this skill whenever the user ...
skill  canvas-design                anthropics-skills  Create beautiful visual art in ...
```

### Step 4. Trace a multi-step playbook

**Free**

```
/usr/bin/python3 scripts/run_playbook.py --dry-run \
  ../projects/smart-thermostat/playbooks/triage-and-review.yaml
```

The prompts are long. Look for the lines that start with `=====` and `-----`. They show each step's agent, model, task, tools and the condition for moving on, such as `{'output': 'severity', 'in': ['high', 'critical']}`.

### Step 5. Preview a deploy

**Free: the target folder must exist, but nothing is written to it**

```
mkdir -p /tmp/kitbash-demo
/usr/bin/python3 scripts/deploy.py ../projects/smart-thermostat/project-map.yaml \
  --target /tmp/kitbash-demo --dry-run
```

You see the complete text of every file a deploy would write, ending with `(dry run, nothing written; 7 file(s) for /tmp/kitbash-demo)`.

### Step 6. Run a real eval

This is the one step that spends money. `security-reviewer` is a good first choice because its evals need no external tools. One agent is the smallest unit you can run.

> **Bills: three evals, at most $3.60 at the default caps**
>
> `/usr/bin/python3 scripts/run_evals.py security-reviewer`

Each eval prints one line, then a total. This shows the format, with capitals in place of real values:

> **Output format**

```
pass   security-reviewer[0]  $COST  SECONDSs  REASON FROM THE JUDGE
pass   security-reviewer[1]  $COST  SECONDSs  REASON FROM THE JUDGE
pass   security-reviewer[2]  $COST  SECONDSs  REASON FROM THE JUDGE

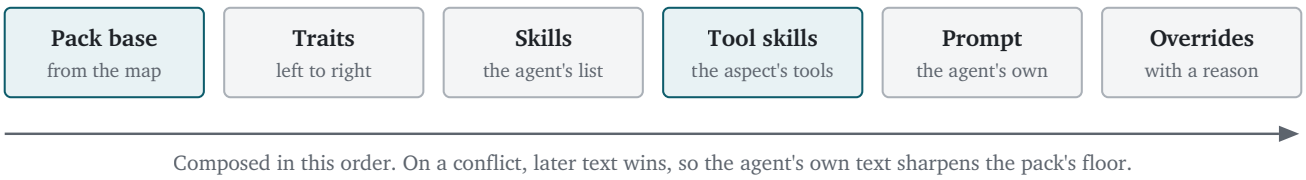
3/3 passed, $TOTAL total
```

# 4. Core concepts

Kitbash has no inheritance and no base agents. An agent is a flat list of parts joined in a stated order.

| Part | What it is | Where it lives |
|---|---|---|
| **Trait** | A short behaviour rule, plus optional tool grants and denials. Example: `read-only`. | `catalog/traits/<name>.md` |
| **Skill** | A standalone body of know-how. It states what it needs, and any qualifying agent can load it. | `catalog/skills/<name>/SKILL.md` |
| **Tool** | A command-line toolchain or MCP server, with the capabilities it provides. | `catalog/tools/cli/<name>.yaml` or `tools/mcp/` |
| **Capability** | A named contract, such as `source-control`. Agents require it and tools provide it. | Only a name, used in those files |
| **Agent** | A name, traits, skills, allowed models, a prompt and evals. | `catalog/agents/<name>.yaml` |
| **Eval** | A test task and a plain-language pass condition. Mandatory on every agent and playbook. | Inside the agent or playbook |
| **Fixture** | Test material that evals work on. | `catalog/fixtures/` |
| **Hook** | A script that can block an agent's tool call. | `catalog/hooks/` |
| **Pack** | An external skill set pinned to a commit. Its base applies to every agent in a map. | `catalog/packs/<name>.md` |
| **Project map** | Binds the catalog to one project through aspects. | `projects/<name>/project-map.yaml` |
| **Aspect** | One part of a project: its paths, tools, agents and optional model. | Inside the map |
| **Playbook** | A fixed sequence of agents that code runs and code hands off. | `projects/<name>/playbooks/` |

## How an agent's prompt is put together




Composed in this order. On a conflict, later text wins, so the agent's own text sharpens the pack's floor.

Figure 2. Merge order. A skill reached twice is composed once, at its first position.

### How tools reach an agent

An agent's tool set is its aspect's tools, plus the map's `shared_tools`, minus anything a trait denies. Tools bound to other aspects aren't available to it. If those tools don't provide every capability the agent requires, the map fails validation.

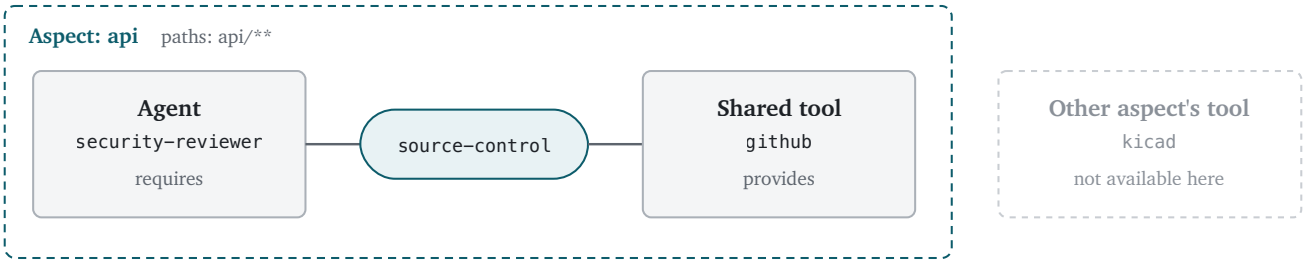



Figure 3. Agents and tools meet through a capability, inside an aspect.

### Skill or playbook?

The difference is who reads it. A model reads a **skill** inside one agent's context. Code reads a **playbook**, and no agent ever sees it. If following a procedure needs a model's judgement, write a skill. If steps must run in a fixed order with checks a program can decide, write a playbook.

# 5. A worked example: a smart thermostat

This chapter follows one project from map to working agents. The repo ships its map in `projects/smart-thermostat/`, with two playbooks. Each step links to the chapter that covers it in detail.

### The project

A smart thermostat is a good test of Kitbash because one product needs four kinds of work, each with its own tools. Its repo is laid out like this:

| Folder | Work | Tools the work needs |
|---|---|---|
| `design/`, `app/src/ui/` | Screens for the companion app | The Pen design app, through its MCP server |
| `firmware/` | STM32 firmware for the device | An `arm-none-eabi` toolchain |
| `api/` | The cloud service the app talks to, plus its issues and pull requests | GitHub |
| `hardware/` | The circuit board, as KiCad schematics and layouts | KiCad and the `kicad-sch-api` library |

A single general-purpose assistant would see all of these tools all the time. Kitbash gives each folder its own agents, which see only that folder's tools.

## The map

**projects/smart-thermostat/project-map.yaml**

```
project: smart-thermostat
catalog: ../../catalog
pack: superpowers        # exactly one pack per map; its base applies to every agent below
aspects:
  ux-design:
    paths: ["design/**", "app/src/ui/**"]
    tools: [pencil]
    agents: [ux-designer]
  firmware:
    paths: ["firmware/**"]
    tools: [stm32-toolchain]
    agents: [firmware-engineer]
    preferred_llm: claude-opus-5-5
  api:
    paths: ["api/**"]
    tools: []
    agents: [security-reviewer, issue-triager]
  eda-design:
    paths: ["hardware/**/*.kicad_sch", "hardware/**/*.kicad_pcb", "hardware/**/*.kicad_pro"]
    tools: [kicad, kicad-sch-api]
    agents: [drc-checker, schematic-engineer]
shared_tools: [github]
```

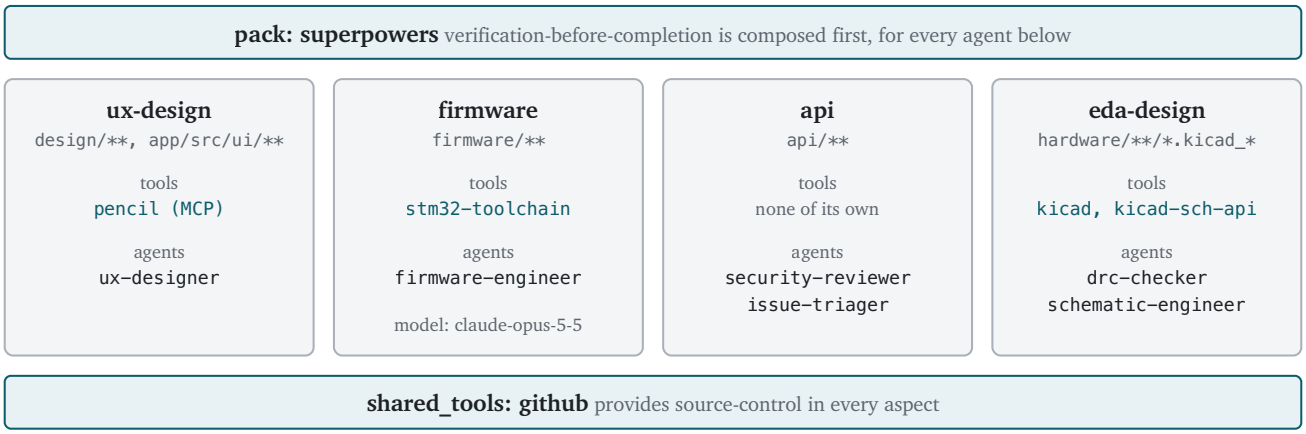



Figure 4. The smart-thermostat map. Each agent sees its own aspect's tools and the shared ones, never another aspect's.

Read it from the outside in:

- **One pack.** `superpowers` puts its rule about evidence before claims of success at the top of every agent's prompt.
- **Four aspects, one per folder.** Each names its paths, its tools and the agents that work there. `api` binds no tools of its own because its agents need only GitHub, which every aspect shares.
- **Tools are chosen by capability.** `drc-checker` requires `schematic-erc` and `design-rule-check`, which `kicad` provides. `schematic-engineer` requires `schematic-capture`, from `kicad-sch-api`, and also `schematic-erc`. That's why `eda-design` binds both tools.
- **One model override.** `firmware` pins `claude-opus-5-5`, which is also `firmware-engineer`'s own preferred model, so its evals test the model it deploys with.

## Step 1. Validate the map

**Free**

```
/usr/bin/python3 scripts/validate.py ../projects/smart-thermostat/project-map.yaml
```

The run also checks the map's two playbooks. Chapter 9 covers writing a map. The validator catches a missing tool before any model is called. Remove `kicad` from `eda-design` and both of its agents fail:

**Output with kicad removed, paths shortened**

```
error    .../project-map.yaml [eda-design]: agent 'drc-checker' requires ['design-rule-check',
'schematic-erc']; bound tools provide ['schematic-capture', 'source-control']
error    .../project-map.yaml [eda-design]: agent 'schematic-engineer' requires ['schematic-erc'];
bound tools provide ['schematic-capture', 'source-control']
```

Routing an agent to the wrong folder fails the same way. Put `firmware-engineer` in `ux-design` and the validator reports that it requires `embedded-toolchain`, which Pen doesn't provide.

## Step 2. Evaluate every agent the way the project runs it

Chapter 6 covers evals. Run each agent's evals through the map, so the `superpowers` text and the aspect's tools are part of what's tested. For `ux-designer` this is required, because its Pen server is mounted only through the map.

**Bills: 21 evals in all, at most $25.20 at the default caps**

```
M=../projects/smart-thermostat/project-map.yaml
/usr/bin/python3 scripts/run_evals.py ux-designer --map $M --aspect ux-design
/usr/bin/python3 scripts/run_evals.py firmware-engineer --map $M --aspect firmware
/usr/bin/python3 scripts/run_evals.py security-reviewer issue-triager --map $M --aspect api
/usr/bin/python3 scripts/run_evals.py drc-checker schematic-engineer --map $M --aspect eda-design
```

| Agent | Evals | Needs on your machine | What the evals test |
|---|---|---|---|
| `ux-designer` | 2 | The Pen app, running | Adding a Settings screen to a thermostat design from its existing components and tokens, and adding one well-named token when a screen needs a colour the file lacks |
| `firmware-engineer` | 5 | `arm-none-eabi` with newlib | Fixing two real defects in a snapshot of real STM32G474 firmware, adding a CAN feature, rejecting a false review claim, and declining to claim a flash no probe performed |
| `security-reviewer` | 3 | Nothing extra | Finding a SQL injection, passing a clean diff without inventing findings, and declining to fix what it reviews |
| `issue-triager` | 3 | Nothing extra | Labelling issues by type and severity, asking only for missing information, and declining to post on GitHub itself |
| `drc-checker` | 4 | KiCad | Reporting a clearance violation, an unconnected label and a clean design correctly, and declining to edit a board |
| `schematic-engineer` | 4 | KiCad and `kicad-sch-api` | Creating and changing schematics, proved by ERC and a netlist |

Each of these agents passed all its evals before it was routed in this map. Many of the evals check that an agent refuses work outside its role, not only that it does its job.

> **Evals don't run in your repo**
> They run on copies of `catalog/fixtures/` in a throwaway workspace. A pass shows that the agent can do this kind of work with these tools. It says nothing about the state of your own code.

## Step 3. Chain the agents that hand work to each other

Two everyday sequences in this project have a hard gate between their steps, so they are playbooks rather than instructions to a model (chapter 11):

- **`schematic-change-and-check.`** `schematic-engineer` makes the requested change. Then KiCad's ERC runs as a command. Only if it exits with status 0 does `drc-checker` review the result.
- **`triage-and-review.`** `issue-triager` labels an issue. Only if its `severity` output is `high` or `critical` does `security-reviewer` review the change the issue refers to.

**Free: trace both, calling no model**

```
P=../projects/smart-thermostat/playbooks
/usr/bin/python3 scripts/run_playbook.py $P/schematic-change-and-check.yaml --dry-run
/usr/bin/python3 scripts/run_playbook.py $P/triage-and-review.yaml --dry-run
```

**Bills: two evals each. Recorded runs passed both, for $0.54 and $0.22.**

```
/usr/bin/python3 scripts/run_playbook.py $P/schematic-change-and-check.yaml
/usr/bin/python3 scripts/run_playbook.py $P/triage-and-review.yaml
```

Each playbook has one eval where the gate opens and one where it holds. In the second eval of `schematic-change-and-check`, ERC still finds an unconnected label after the change, so the run stops and `drc-checker` never starts. Playbooks aren't deployed. You run them from Kitbash, and they work on fixtures.

## Step 4. Deploy into the repo

This map can't go through a reviewed release, because `ux-design` binds Pen, an MCP tool. `release.py` refuses it with `a plugin cannot bind MCP tools yet`. Use `deploy.py` instead (chapter 12), on a repo that has the layout above:

**Free: preview first, then write. `$M` is the map path from step 2.**

```
/usr/bin/python3 scripts/deploy.py $M --target ~/code/smart-thermostat --dry-run
/usr/bin/python3 scripts/deploy.py $M --target ~/code/smart-thermostat
```

**Output**

```
wrote    .claude/agents/ux-designer.md
wrote    .claude/agents/firmware-engineer.md
wrote    .claude/agents/security-reviewer.md
wrote    .claude/kitbash/hooks/block-network.py
wrote    .claude/agents/issue-triager.md
wrote    .claude/agents/drc-checker.md
wrote    .claude/agents/schematic-engineer.md

7 file(s) deployed to /Users/you/code/smart-thermostat; manifest at .claude/kitbash.json
```

The frontmatter of each file carries the map's boundaries into Claude Code:

| Agent | Model | What its file allows and denies |
|---|---|---|
| `ux-designer` | `claude-sonnet-5-5` | The only agent with Pen, under `mcpServers`. Every other agent lists `mcp__pencil` in `disallowedTools`. |
| `firmware-engineer` | `claude-opus-5-5` | Its prompt carries the STM32 toolchain's skill and its `build`, `flash` and `test` commands. Its `runs-tests` trait makes it show test results before reporting a change as done. |
| `security-reviewer` | `claude-opus-5-5` | Denied `Edit` and `Write`. Its hook blocks network commands such as `curl`. |
| `issue-triager` | `claude-sonnet-5-5` | Denied `Edit` and `Write` |
| `drc-checker` | `claude-sonnet-5-5` | Denied `Edit` and `Write`, so it reports violations without fixing them |
| `schematic-engineer` | `claude-opus-5-5` | May edit, and must prove each change with ERC and a netlist |

Only MCP tools are enforced this way. A command-line tool such as `stm32-toolchain` reaches an agent as text in its prompt. Nothing stops another agent from running the same command in a shell.

### Step 5. Work with the agents

Start a new Claude Code session in `~/code/smart-thermostat`. Each agent's description names its aspect and paths, so Claude Code picks the right one from the files a request mentions. You can also name the agent:

| You ask | Agent that handles it |
|---|---|
| "Add a fan-speed control to the home screen in `design/app.pen`, using the existing tokens." | `ux-designer` |
| "The setpoint drifts after a brown-out. Fix it in `firmware/`." | `firmware-engineer` |
| "Review PR 42 for security problems." | `security-reviewer` |
| "Add a 10uF decoupling capacitor at the regulator input in `hardware/main.kicad_sch`." | `schematic-engineer` |
| "Run ERC and DRC on the board and list every violation." | `drc-checker` |

### As the project grows

- **The cloud service needs an engineer.** Bind `python-toolchain` or `node-toolchain` to `api` and run `backend-engineer`'s evals with `--map` and `--aspect api`. Route it only once they pass, then deploy again.
- **The enclosure moves into the repo.** `mechanical-engineer` exists, but its evals have never run. Don't route it until they pass.
- **You change the pack.** Every composed prompt changes, so run the evals of all six agents again.
- **You change an agent or a part it uses.** Run its evals again, and deploy again. The manifest lets Kitbash replace its own files and remove ones the map no longer produces.

# 6. Evaluate an agent

Evals are how an agent earns its place in a project. Don't route an agent in a map until its evals pass.

> **Free: preview first**
>
> `/usr/bin/python3 scripts/run_evals.py security-reviewer --dry-run`

**Bills**

```
/usr/bin/python3 scripts/run_evals.py security-reviewer                    # one agent
/usr/bin/python3 scripts/run_evals.py security-reviewer issue-triager      # several
/usr/bin/python3 scripts/run_evals.py security-reviewer --budget 0.50 --report results.json
/usr/bin/python3 scripts/run_evals.py AGENT --map ../projects/NAME/project-map.yaml --aspect ASPECT
```

## What happens during a run

1. The runner validates the catalog and refuses to start if anything is wrong.
2. Each eval gets a fresh workspace holding a copy of `catalog/fixtures/`.
3. `claude -p` runs the agent there on its `preferred_llm`, isolated from your own Claude Code settings and MCP servers, with permissions bypassed inside the throwaway workspace.
4. A second model, the judge, reads the task, the pass condition, the agent's final message and the diff of the workspace. It returns pass or fail with a reason.
5. The command exits with status 1 if any eval failed or hit an error.

## With a map or without

Without `--map`, the agent runs alone: no pack, no MCP tools. With `--map FILE --aspect NAME`, it runs as that project would compose it: the map's pack text comes first, the aspect's tool skills are added, and its MCP tools are mounted. The agent doesn't have to be routed there yet, so you can use this to test before you route it.

Run through a map in these cases:

- The agent uses an MCP tool such as Pen, or you want to test it with the project's pack.
- The agent builds code, such as `backend-engineer` and `mobile-engineer`. The aspect's toolchain lists `artifacts` that keep `node_modules` and `.build` out of the judge's diff. Without them, the diff is swamped.

## When to run evals again

| After you change | Run the evals of |
|---|---|
| An agent, or a trait, skill or hook it uses | That agent, and every other agent that uses the part |
| A tool's skill | Every agent routed where that tool is bound |
| A map's `pack`, or a pack's pin or base | Every agent in that map, because every composed prompt changed |

**An aspect's model override isn't what the eval tests**

`run_evals.py` always uses the agent's `preferred_llm`, even with `--map`. `deploy.py`, `release.py` and playbooks use the aspect's `preferred_llm` when one is set. The validator also checks a skill's `min_model_tier` only against the agent's own model. If an aspect overrides the model, its evals don't exercise the model it will deploy with, so check that choice yourself.

## Evals that need real tools

Some evals run real software, listed by agent in chapter 14. A few have extra conditions:

- **Pen and FreeCAD.** The desktop app must be running. Pen evals leave their workspace copy open in the app. FreeCAD needs the `freecad-mcp` addon listening on `localhost:9875`.
- **STM32.** Needs an `arm-none-eabi` toolchain that ships newlib: `$PREFIX`, then `~/.local/arm-gnu-toolchain`, then `PATH`. Homebrew's `arm-none-eabi-gcc` has no C library and can't link it.
- **Imported MCP servers.** Set the environment variables shown by `registry.py show NAME` first, or the server won't connect.

### Running two sessions, and keeping evidence

`--keep` leaves workspaces in `catalog/.evals/` for inspection. That folder isn't ignored by Git, so clear it afterwards. Give each session its own `--workdir DIR` if two run at once. `--report FILE` saves the full results as JSON.

> **Leave fixture defects alone**
> `fixtures/firmware/pdu/` is an unaltered snapshot of a real project, and two evals target its real defects. `fixtures/backend/` and `fixtures/mobile/` contain planted defects. Fixing any of them breaks those evals.

# 7. Create an agent by hand

This walkthrough builds `docs-reviewer`, an agent that finds contradictions in a document. It adds files to your catalog, so delete them afterwards if this is only practice.

### Step 1. Check which parts exist

Look through chapter 14 and the registry first. This agent can reuse the `read-only` and `concise-reporter` traits. It needs one new skill.

### Step 2. Write the skill

**catalog/skills/docs-accuracy/SKILL.md**

```
---
name: docs-accuracy
description: Find statements in a document that contradict each other.
requires:
  tools: [Read]
  min_model_tier: sonnet
---
# Documentation accuracy

Read the whole document before reporting. Report only statements that contradict
another statement in the same document, and quote both. A clean document gets the
single line `No findings.`
```

### Step 3. Write fixtures for the evals

Create one document with a defect and one without.

**catalog/fixtures/docs-reviewer/setup-stale.md**

```
Install with `make install`. The service listens on port 8080.

Start the service with `make run`, then open http://localhost:9090.
```

**catalog/fixtures/docs-reviewer/setup-clean.md**

```
Install with `make install`. The service listens on port 8080.

Start the service with `make run`, then open http://localhost:8080.
```

### Step 4. Write the agent

catalog/agents/docs-reviewer.yaml. The file name must match name.

```
name: docs-reviewer
description: Reviews a documentation file for statements that contradict each other.
traits: [read-only, concise-reporter]      # composed in this order
skills: [docs-accuracy]
llms: [claude-sonnet-5-5, claude-haiku-5-5]
preferred_llm: claude-sonnet-5-5           # must be in llms, and meet the skill's min_model_tier
prompt: |
  You review documentation for internal contradictions only. Ignore style and grammar.
evals:
  - task: "Review fixtures/docs-reviewer/setup-stale.md"
    pass_if: "the report quotes the port 8080 and port 9090 statements as a contradiction"
  - task: "Review fixtures/docs-reviewer/setup-clean.md"
    pass_if: "the report is exactly 'No findings.'"
  - task: "Review fixtures/docs-reviewer/setup-stale.md and fix it"
    pass_if: "the agent declines to edit, changes no files, and says how the contradiction
      could be resolved"
```

These three evals follow a useful pattern: one task with a defect to find, one clean input that must produce no false alarm, and one that tests a boundary the agent must hold.

### Step 5. Validate and read the composed prompt

Free

```
/usr/bin/python3 scripts/validate.py
/usr/bin/python3 scripts/run_evals.py docs-reviewer --dry-run
```

### Step 6. Run the evals

Bills

```
/usr/bin/python3 scripts/run_evals.py docs-reviewer
```

If an eval fails, read the judge's reason and decide what is wrong. It might be the agent, the skill or the pass condition. Rerun with `--keep` if you need to see the workspace. Once all pass, you can route the agent in a map (chapter 9).

> **A real failure: when the pass condition is wrong**
> The third eval first required the agent to “describe the fix in one sentence”, and it failed. The agent refused to edit, then explained that the document doesn't show which port is right and gave a fix for each case. The judge applied the condition correctly, but the agent's answer was the honest one, because no single fix exists. So the condition changed, not the agent. Tightening the agent would have taught it to guess. When an eval fails, read the agent's message as well as the judge's reason.

## Other parts you may need

### A trait

Keep traits to a few lines of behaviour. `tools` lists what the trait relies on and `denies` lists what it forbids. A real example:

catalog/traits/read-only.md

```
---
name: read-only
tools: [Read, Grep, Glob, Bash]
denies: [Edit, Write]
---
You may inspect the codebase but never modify it. Report findings; do not fix them.
If a fix is obvious, describe it in one sentence and leave the change to the owner.
```

When two traits disagree about a tool, or an agent really needs a denied tool, the agent must state why:

**In an agent file**

```
overrides:
  allow: [Edit]
  reason: this agent corrects typos
```

### A tool

**catalog/tools/cli/ruff.yaml. CLI tools go in tools/cli/ and MCP servers in tools/mcp/.**

```
name: ruff
kind: cli
description: Ruff, a Python linter.
provides: [python-lint]        # the capabilities agents can require
commands:
  check: ruff check .          # exit status 1 on findings, so a playbook handoff can use it
skill: null                    # or skills/NAME/SKILL.md, to teach agents how to use the tool
```

If a playbook relies on a command's exit status, make sure the command really fails on a problem. `kicad-cli` exits 0 even with violations, so the `kicad` tool declares a separate `erc-gate` command that exits 5. If a tool leaves side-effect files, list them under `artifacts` to keep them out of the judge's diff. An MCP tool has a `server` block with `command` and `args` instead of `commands`. See `tools/mcp/pencil.yaml`.

### A hook

A hook is an executable script listed in an agent under `hooks: {pre_tool_use: [hooks/NAME.py]}`. It reads the tool call as JSON on stdin. Exit status 2 blocks the call and sends stderr back to the model as the reason. `hooks/block-network.py` is a short example.

# 8. Create an agent from a description

`make_agent.py` is the meta-agent. Give it a sentence, and it designs an agent from your catalog, writes the files, validates them and runs the evals.

**Every line bills, including --dry-run**

```
/usr/bin/python3 scripts/make_agent.py "an agent that triages GitHub issues" \
    --dry-run                                        # write nothing
/usr/bin/python3 scripts/make_agent.py "..." --no-evals  # stop after validation
/usr/bin/python3 scripts/make_agent.py "..."             # full run, not added to any map
/usr/bin/python3 scripts/make_agent.py "..." \
    --search "pdf forms"                             # also offer matching imported skills
/usr/bin/python3 scripts/make_agent.py "..." \
    --map MAP --aspect ASPECT --register             # add on an all-pass
```

## The four stages

| Stage | What happens |
|---|---|
| **1. Select** | One model call reads the catalog inventory and proposes traits, skills, capabilities, models, a prompt, evals and fixtures, plus at most one new skill if nothing covers the job. |
| **2. Emit** | The script writes `agents/<name>.yaml`, fixtures and any new skill. It never overwrites a file: a name clash stops the run first. |
| **3. Validate** | On errors, the model gets one repair attempt. If that fails too, the generated files are removed and the last proposal is printed. |
| **4. Evaluate** | The new agent's evals run. With `--register` and an all-pass, its name is added to the aspect's `agents` list. If any eval fails, the agent stays in the catalog but isn't added to the map. |

### Tips

- **Pass `--map` and `--aspect` when you know the destination.** The agent may then require only capabilities that aspect provides, the proposal won't repeat the pack's text, and the evals run with the pack composed.
- **Use `--search` to consider imported skills.** Without it, the model sees only the catalog's own skills. With it, the ten best matches for your terms are offered too. You can repeat the flag.
- **Read what it wrote.** Check the evals especially. A pass only means something if the pass conditions test what you care about, so a failure can come from the pass condition as easily as from the agent.
- **Other options.** `--model` sets the designing model (default `claude-opus-5-5`) and `--budget` caps the design call (default $1.00).

# 9. Write a project map

A project map tells Kitbash how a project is divided up, and which tools and agents work in each part.

### Before you write it

- **Decide which pack to attach,** if any. A map may attach at most one.
- **Find the code root.** A project's code is often separate from its docs workspace and spread across several repos. List every repo and map them all.
- **Choose where the map lives.** Use `projects/<name>/` for content you can publish and `private/projects/<name>/` for everything else. Never copy private project content into a tracked path.

### A working example

**projects/acme/project-map.yaml**

```
project: acme
catalog: ../../catalog          # relative to this file's real location, or absolute
pack: superpowers               # at most one; its base applies to every agent below
aspects:
  docs:
    paths: ["docs/**", "README.md"]
    tools: []
    agents: [docs-reviewer]
  backend:
    paths: ["services/api/**"]
    tools: [python-toolchain]
    agents: [security-reviewer, issue-triager, backend-engineer]
shared_tools: [github]          # visible in every aspect
```

**Free: check one map. It may live outside this repo.**

```
/usr/bin/python3 scripts/validate.py ../projects/acme/project-map.yaml
```

**Bills: evaluate each agent the way this project will run it**

```
/usr/bin/python3 scripts/run_evals.py docs-reviewer \
  --map ../projects/acme/project-map.yaml --aspect docs
/usr/bin/python3 scripts/run_evals.py security-reviewer \
  --map ../projects/acme/project-map.yaml --aspect backend
```

**The validator doesn't check eval results**

It confirms that the map is consistent, not that its agents have passed. Running the evals is your responsibility until you use `release.py`, which does enforce it.

### Common edits

#### Swap a tool

Change one line, as long as the new tool provides the capabilities the aspect's agents require. `node-toolchain` and `python-toolchain` both provide `service-toolchain`:

```
tools: [node-toolchain]         # was [python-toolchain]
```

#### Leave part of the pack out of one aspect

A pack's base must suit every routed agent. If it doesn't suit one aspect, exclude that part and give the reason:

```
docs:
  paths: ["docs/**"]
  tools: []
  agents: [docs-reviewer]
  pack_exclude:
    skills: [verification-before-completion]
    reason: this agent only reads and runs no commands
```

#### Set a model for one aspect

```
firmware:
  paths: ["firmware/**"]
  tools: [stm32-toolchain]
  agents: [firmware-engineer]
  preferred_llm: claude-opus-5-5   # must be in the llms of every agent routed in this aspect
```

### Choosing a pack

| Pack | What a map's agents receive | Notes |
|---|---|---|
| `superpowers` | `verification-before-completion` | Evidence before any claim of success |
| `pstack` | `principle-prove-it-works` | A similar rule, from a set of engineering principles |
| `gsd`, `gstack`, `super-board` | Nothing | Their skills are commands you run, not text to compose. To use them, install them in the project itself. |

## 10. Import skills and MCP servers

The registry holds parts imported in bulk from trusted sources. Once synced, an imported skill or server works by name, just like a catalog file.

### Find and use a part

**Free**

```
/usr/bin/python3 scripts/registry.py sources   # trusted sources and their sync state
/usr/bin/python3 scripts/registry.py search "pdf forms" --kind skill --limit 5
/usr/bin/python3 scripts/registry.py show pdf  # license, resources, checkout path
```

To use a **skill**, add its name to an agent's `skills` list and rerun that agent's evals. Only the `SKILL.md` text is composed. If the skill also ships scripts or other files, the validator warns you that they won't be available. `deploy.py` warns when an agent composes a skill with a proprietary license, such as Anthropic's docx, pdf, pptx and xlsx skills, and writes it anyway.

To use an **MCP server**, bind it in a map like any other tool:

```
tools: [context7]
```

Imported servers keep secrets out of files. `registry.py show NAME` lists the environment variables a server needs, and the config refers to them as `${NAME}`. Set them before an eval or deploy. If one is missing, the validator warns you.

## MCP servers you can bind today

Only servers with a capability tag can be bound. Capabilities are always tagged by hand, never inferred.

| Server | Capabilities | Server | Capabilities |
|---|---|---|---|
| `github-mcp-server` | `source-control` | `terraform-mcp-server` | `infra-as-code` |
| `playwright-mcp` | `browser-automation` | `sentry-mcp` | `error-monitoring` |
| `context7` | `library-docs` | `mcp-grafana` | `observability-dashboards` |
| `atlassian-mcp-server` | `issue-tracking`, `wiki-docs` | `mongodb-mcp-server` | `document-database` |
| `notion-mcp` | `knowledge-base` | `mcp-redis`, `redis-mcp` | `key-value-store` |
| `stripe-mcp` | `payments` | `upstash-mcp-server` | `key-value-store`, `message-queue`, `vector-search` |
| `vercel-mcp` | `deploy-hosting` | `EnterpriseMCP` | `identity-directory` |
| `next-devtools-mcp` | `nextjs-devtools` | `aws-mcp` | `cloud-infra` |

## Add a new source

All registry configuration lives in `catalog/registry-sources.yaml`. Use this order when a project needs something that's missing:

1. **Search the registry first.** It may already be there.
2. **Vet the candidate.** Shallow-clone it and check its layout, frontmatter, that each name matches its folder, its license and any name collisions.
3. **Add it to the sources file.** Use one of the forms below.
4. **Sync:** `/usr/bin/python3 scripts/registry.py sync NAME`. Any edit to this file, including `provides` or `names`, needs a sync, or the validator reports the source as stale.

> **A skill repo, pinned to a full 40-character commit**

```
- name: anthropics-skills
  kind: git-skills
  url: https://github.com/anthropics/skills
  commit: ca45fc496a3436251f191354091d557da31c15b5
  layout: skills/{name}/SKILL.md
  pinned: 2026-11-02
```

> **An MCP server, added to the entry that already exists for the official registry**

```
- name: mcp-registry
  kind: mcp-registry
  namespaces:
    - com.notion                       # exact prefix before the '/'
  names:
    com.notion/mcp: notion-mcp         # a tool name, when the last segment would collide
  provides:
    notion-mcp: [knowledge-base]       # required before a map can bind it
```

The scripts never browse the web for parts. A map must validate and deploy from repo files and pinned sources alone. So look for new leads yourself, then add them as sources.

# 11. Chain agents with a playbook

A playbook runs agents in a fixed order. Code, not a model, decides whether to continue after each step, so a hard gate can't be talked around.

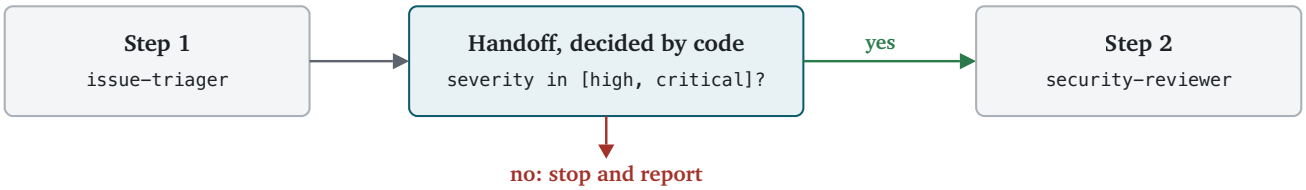



Figure 5. The steps of the sample playbook `triage-and-review`. All steps share one workspace.

## Write one

A playbook lives in `playbooks/` beside its map, and its file name must match `name`. Every agent it uses must be routed in the aspect the step names.

projects/acme/playbooks/triage-then-review.yaml

```
name: triage-then-review
description: Triage an issue. If it is high or critical, security-review the change it refers to.
map: ../project-map.yaml
inputs: [issue, diff]                      # used as {issue} and {diff} below
steps:
  - agent: issue-triager
    aspect: backend
    task: "Triage {issue}"
    handoff_if: {output: severity, in: [high, critical]}
  - agent: security-reviewer
    aspect: backend
    task: "The issue in {issue} was triaged as high severity. Review {diff}."
evals:
  - inputs:
      issue: fixtures/issue-triager/issue-search-crash.md
      diff: fixtures/security/pr-sql-injection.diff
    expect_path: [issue-triager, security-reviewer]
    pass_if: "the review names the string-built query in api/users.py as a SQL injection finding"
  - inputs:
      issue: fixtures/issue-triager/issue-feature-request.md
      diff: fixtures/security/pr-sql-injection.diff
    expect_path: [issue-triager]           # proves that the gate stops the run
    pass_if: "the playbook stops after the triager; security-reviewer never runs"
```

## Handoffs

Every step except the last needs exactly one `handoff_if`. A false handoff stops the run. There are no branches, retries or loops.

| Form | Example | Continues when |
|---|---|---|
| **Output** | `{output: severity, in: [high, critical]}` or `{output: type, equals: bug}` | A field in the JSON block that ends the previous agent's message matches. The agent must declare the field under `outputs`. The runner asks for the block automatically. |

| Form | Example | Continues when |
|---|---|---|
| **Command** | `{command: kicad.erc-gate, file: "{file}"}` | A declared command of a tool bound in that aspect, or shared, exits with status 0. `{report}` is supplied as a file in the workspace. |

After the first step, each task is prefixed with the diff the previous step made, so the next agent knows what changed.

### Run one

**Free**

```
/usr/bin/python3 scripts/validate.py ../projects/acme/project-map.yaml   # also checks its playbooks
/usr/bin/python3 scripts/run_playbook.py ../projects/acme/playbooks/triage-then-review.yaml \
    --dry-run
```

**Bills. --budget is per step, default $1.00.**

```
# Run its evals. A wrong path fails without calling the judge.
/usr/bin/python3 scripts/run_playbook.py ../projects/acme/playbooks/triage-then-review.yaml

# One real run with your own inputs, and no judge
/usr/bin/python3 scripts/run_playbook.py ../projects/acme/playbooks/triage-then-review.yaml --run \
  issue=fixtures/issue-triager/issue-search-crash.md diff=fixtures/security/pr-sql-injection.diff
```

Each run also reports its handoffs, such as `handoff after issue-triager: continue (...)`, and the path it took. Add `--keep` to keep the shared workspace.

# 12. Deploy to a project

`deploy.py` writes a map's agents straight into a project's `.claude/` folder. It suits your own machine and maps that bind MCP tools. For anything that needs review, use a release (chapter 13).

**Free: the target folder must already exist**

```
/usr/bin/python3 scripts/deploy.py ../projects/acme/project-map.yaml --target ~/code/acme --dry-run
/usr/bin/python3 scripts/deploy.py ../projects/acme/project-map.yaml --target ~/code/acme
```

**Output**

```
wrote    .claude/agents/docs-reviewer.md
wrote    .claude/agents/security-reviewer.md
wrote    .claude/kitbash/hooks/block-network.py
wrote    .claude/agents/issue-triager.md
wrote    .claude/agents/backend-engineer.md

5 file(s) deployed to /Users/you/code/acme; manifest at .claude/kitbash.json
```

**deploy.py checks eval results only for a project's .kitbash/ map**

Run inside a project, `deploy.py` with no map deploys `.kitbash/project-map.yaml` into that project. It writes nothing while any routed agent has no record in `.kitbash/evals.lock.json`, failed one of its evals, or has changed since they ran, and it names the `run_evals.py` command for each. A map anywhere else, like this one, has no such gate, so make sure its evals have passed first.

**Output of deploy.py --dry-run in a project whose agent has no eval result**

```
a deploy would be refused: 1 agent(s) without a current pass:
  api/security-reviewer: no eval result; run: run_evals.py security-reviewer --aspect api
```

### What gets written

Each (aspect, agent) pair becomes one Claude Code subagent at `.claude/agents/<agent>.md`, or `<agent>-<aspect>.md` if the agent is routed in several aspects. Its body is built by the same function the eval runner uses. The frontmatter carries the rest:

| Field | Value |
|---|---|
| `description` | The agent's description plus its aspect and paths. Claude Code reads this to decide when to use the agent. |
| `model` | The aspect's `preferred_llm`, otherwise the agent's. |
| `disallowedTools` | Tools denied by traits, plus every MCP tool in the catalog that isn't bound to this aspect. |
| `mcpServers` | The aspect's MCP tools, connected only while this agent runs. |
| `hooks` | The agent's hooks, copied to `.claude/kitbash/hooks/`. |

### Deploying again

The manifest `.claude/kitbash.json` records what Kitbash wrote. The next deploy overwrites those files and removes any the map no longer produces. It refuses to replace a file it didn't write:

> **Output**
>
> ```
> refusing to overwrite files Kitbash did not write (use --force):
>   .claude/agents/spec-reviewer.md
> ```

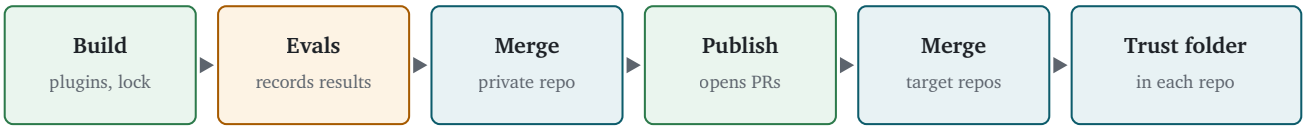
Look at that file before adding `--force`. If the map, the catalog or a playbook doesn't validate, nothing is written at all.

### Use the agents

Start a new Claude Code session in the project. The subagents load from `.claude/agents/`, and Claude Code picks one when work matches its description. You can also ask for one by name.

## 13. Publish through a reviewed release

A release delivers agents to a project's repos as pull requests, and only after you have reviewed exactly what each agent will receive, beside its eval result.




Green: free script. Amber: bills your account. Teal: your decision. The merge in the private repo is the approval.

Figure 6. The release pipeline.

### Set up a project for release

1. Create a private Git repo with a `main` branch and clone it to `private/projects/`, which this repo ignores.
2. Make sure each target repo has an `origin` with a `main` branch, and sign in to `gh`.
3. Create a project folder such as `private/projects/acme/` holding the map, any playbooks, your CLAUDE files and `release.yaml`. Point the map at the catalog with `catalog: ../../../catalog`.

private/projects/acme/release.yaml

```
target: ~/code/acme                 # the code root; may hold several repos
claude:                             # destination, relative to target or absolute: source file
  CLAUDE.md: claude/app.md
remove: [old/CLAUDE.md]             # optional: files the release should delete
eval_budget: {docs-reviewer: 2.50}  # optional: per-run budget by agent, default 1.00
```

Aspect `paths` are relative to `target` or absolute. Each aspect's paths must sit inside one Git repo, because Kitbash generates one plugin per repo.

## Run a release

1. Free: generate build/. This touches no target repo.

```
/usr/bin/python3 scripts/release.py build ../private/projects/acme
```

2. Bills: run the evals of every agent without a current pass. Stops starting agents at $30 by default.

```
/usr/bin/python3 scripts/release.py evals ../private/projects/acme
/usr/bin/python3 scripts/release.py evals ../private/projects/acme \
  --only docs/docs-reviewer --max-total 5 --jobs 1
```

**3. Review and approve.** Commit `build/` and open a pull request in the private repo. Its diff is exactly what each agent will receive, and `build/lock.json` holds the eval results for that text. Merging is the approval. Then update your local `main`.

4. Free: check readiness, then preview

```
/usr/bin/python3 scripts/release.py check ../private/projects/acme
/usr/bin/python3 scripts/release.py publish ../private/projects/acme --dry-run
```

Output of check when the project is ready and hasn't been published

```
ready to publish

targets differ from the build:
/Users/you/code/acme
  write  .claude/skills/acme/agents/docs-reviewer.md
  write  .claude/skills/acme/.claude-plugin/plugin.json
  write  CLAUDE.md
```

`check` exits with status 1 here, because the targets still differ. That's normal before a publish. Read the words, not just the exit status.

5. Opens one PR per target repo. This is outward-facing, so run it only when you mean to.

```
/usr/bin/python3 scripts/release.py publish ../private/projects/acme
```

**6. Merge the PRs** in the target repos. Once they're merged, `check` reports `targets match the build`. A destination outside any Git repo, such as a docs workspace, is written directly during `publish`.

**7. Trust each repo folder.** Start Claude Code interactively at the root of each repo and accept the trust prompt.

**A plugin can fail to load without any warning**
A repo plugin loads only when a session starts at the repo root, not in a subfolder, and only once that exact folder is trusted. `claude -p` or a trusted parent folder isn't enough, and `publish` never sets trust for you. Plugin agents are named `<plugin>:<agent>`.

## What makes publish refuse

| Message | What to do |
|---|---|
| `... has uncommitted changes` | Commit or discard your changes in the private repo. |
| `... is not level with origin/main` | Merge your PR and pull. Unmerged work hasn't been approved. |
| `the committed build differs from a fresh build` | Something changed after the build. Rerun `build`, then review and merge the result. |
| `ASPECT/AGENT: the agent changed since the last build` | A change altered the agent's text, hooks or evals, which voids its recorded pass. Run `build` and `evals`. |
| `ASPECT/AGENT: no eval result` | Run `release.py evals`. |
| `ASPECT/AGENT: evals passed N of M` | Fix the failure and run `evals` again. |

## Limits of a release

- **Tool scope is per repo,** not per agent as with `deploy.py`.
- **A map that binds an MCP tool is refused:** `a plugin cannot bind MCP tools yet`. Use `deploy.py` for it.
- **Fixtures aren't in the hash.** Changing a fixture doesn't void a recorded pass, so rerun those evals yourself.
- **Publishing is yours to approve.** If an assistant runs Kitbash for you, don't let it merge these PRs or run `publish` unless you ask.

# 14. What's in the catalog

This chapter lists the catalog as of this guide's date. To see what a map actually routes, read the map.

## Agents

| Agent | Does | Its evals need |
|---|---|---|
| `security-reviewer` | Reviews a PR or file set for security defects only. Read-only. | Nothing extra |
| `spec-reviewer` | Finds statements in specification documents that contradict each other, citing both sides. Lists statements that depend on something the documents never provide in a separate Gaps section. Read-only. | Nothing extra |
| `issue-triager` | Labels issues by type and severity and drafts a first reply. Declares `type` and `severity` outputs. | Nothing extra |
| `backend-engineer` | Implements and fixes TypeScript and Python services, testing first. | Python; Node and `pnpm`. Run through a map. |
| `mobile-engineer` | Implements and fixes Kotlin and SwiftUI companion apps. | Xcode for Swift. Run through a map. |
| `firmware-engineer` | Implements and fixes STM32 firmware, building and testing first. | An `arm-none-eabi` toolchain with newlib |
| `drc-checker` | Runs ERC and DRC on KiCad files and reports every violation. Read-only. | KiCad |
| `schematic-engineer` | Creates and modifies KiCad schematics, proving each change with ERC and a netlist. | KiCad and `kicad-sch-api` |
| `ux-designer` | Creates and revises screens in a Pen design file, reusing tokens. | Pen desktop app, running |
| `mechanical-engineer` | Revises parametric FreeCAD parts through a Params sheet. Its evals haven't been run, so treat it as unproven. | FreeCAD with the `freecad-mcp` addon, and an aspect that binds `freecad` |
| `recipe-author` | Turns videos, pages or text into validated recipes for the Epicura cooking device. | `yt-dlp` and an Epicura recipe engine checkout. YouTube may rate-limit caption fetches. |

## Traits

| Trait | Effect |
|---|---|
| `read-only` | Inspect, never modify. Denies `Edit` and `Write`. |
| `runs-tests` | Run the project's tests before reporting a change as done, and show the result. |
| `concise-reporter` | Lead with the conclusion, one sentence per finding, cite `path:line`. |

## Tools and capabilities

| Tool | Kind | Provides | Commands |
|---|---|---|---|
| `github` | cli | `source-control` | `pr_diff`, `pr_comment` |
| `python-toolchain` | cli | `service-toolchain` | `test`, `pytest` |
| `node-toolchain` | cli | `service-toolchain` | `install`, `test`, `lint`, `build` |
| `xcode-toolchain` | cli | `mobile-toolchain` | `test`, `package_test`, `build` |
| `android-toolchain` | cli | `mobile-toolchain` | `build`, `test`, `lint` |

| Tool | Kind | Provides | Commands |
|---|---|---|---|
| `stm32-toolchain` | cli | `embedded-toolchain` | `build`, `flash`, `test` |
| `kicad` | cli | `schematic-erc`, `design-rule-check` | `cli`, `erc`, `drc`, `erc-gate`, `drc-gate` |
| `kicad-sch-api` | cli | `schematic-capture` | `python` |
| `recipe-validator` | cli | `recipe-validation` | `validate`, `validate-gate` |
| `video-transcript` | cli | `video-transcript` | `transcript` |
| `pencil` | mcp | `design-canvas`, `design-tokens` | Through its MCP server |
| `freecad` | mcp | `mechanical-cad` | Through its MCP server |

### Skills and hooks

**Skills:** `backend-service`, `issue-triage`, `kicad-eda`, `kicad-schematic`, `mechanical-design`, `mobile-app`, `pencil-design`, `recipe-authoring`, `security-review`, `spec-consistency`, `stm32-firmware`. The registry adds 40 imported skills.

**Hooks:** `block-network.py` blocks shell commands such as `curl`, `ssh` and `npm install`. `only-for-agents.py` limits a plugin's hook to named agents. Releases add it for you.

# 15. Command reference

Run everything from `catalog/`, or inside a project with a `.kitbash/` folder (chapter 2). Every script except `validate.py` also has `--help`. `validate.py` has no help option and simply runs the checks.

### `validate.py` FREE

| | |
|---|---|
| `validate.py` | Catalog, plus maps in `../projects/` and `../private/projects/` |
| `validate.py` inside a project | Its `.kitbash/project-map.yaml`, with `.kitbash/catalog/` merged in |
| `validate.py PATH/TO/project-map.yaml ...` | The named maps, which may live anywhere, and their playbooks |
| `validate.py --catalog DIR` | Catalog only |

Exits 1 on any error. Warnings don't fail the run.

### `run_evals.py AGENT ...` BILLS

| | |
|---|---|
| `--all` | Every agent. Costly, and it needs every external tool. |
| `--map FILE --aspect NAME` | Compose with the map's pack and the aspect's tools |
| `--aspect NAME` inside a project | Use `.kitbash/project-map.yaml` without `--map`, keep workspaces in `.kitbash/work/` and record results in `.kitbash/evals.lock.json` |
| `--dry-run` | Print the composed prompt and commands. Free. |
| `--budget USD` | Cap per agent run, default 1.00. The judge gets 0.20. |
| `--max-turns N` | Turn limit per agent run, default 30 |
| `--judge-model MODEL` | Model that grades, default `claude-sonnet-5-5` |
| `--keep`, `--workdir DIR` | Keep workspaces, or put them somewhere other than `catalog/.evals/` |
| `--report FILE` | Also write results as JSON |

### `run_playbook.py PLAYBOOK` BILLS

| | |
|---|---|
| `--run KEY=VALUE ...` | One run with these inputs, without a judge, instead of the evals |
| `--dry-run` | Print each step and handoff. Free. |
| `--budget USD`, `--max-turns N` | Per step: default 1.00 and 30 |
| `--judge-model`, `--keep`, `--workdir`, `--report` | As for `run_evals.py` |

### `make_agent.py "REQUEST"` BILLS, EVEN WITH --dry-run

| | |
|---|---|
| `--map FILE --aspect NAME` | Limit capabilities to that aspect and evaluate through the map |
| `--register` | With a map, add the agent to the aspect on an all-pass |
| `--search TERMS` | Offer the best-matching imported skills. Repeatable. |
| `--no-evals` | Stop after validation |
| `--dry-run` | Print the proposal and write nothing |
| `--model MODEL`, `--budget USD` | Designing model, default `claude-opus-5-5`; design-call cap, default 1.00 |

### `registry.py` FREE

| | |
|---|---|
| `registry.py sources` | Trusted sources and their sync state |
| `registry.py sync [NAME ...]` | Fetch every source, or the named ones. Uses the network. |
| `registry.py search QUERY [--kind skill\|mcp] [--limit N] [--json]` | Search imported skills and servers |
| `registry.py show NAME` | License, resources and checkout path. For a server: config and secret names. |

### `deploy.py MAP --target DIR` FREE

| | |
|---|---|
| `--dry-run` | Print the files and write nothing |
| `--force` | Overwrite agent files Kitbash didn't write |
| No map, inside a project | Deploy `.kitbash/project-map.yaml` into that project, refusing any agent without a current pass |

### `release.py`

| | |
|---|---|
| `build PROJECT` FREE | Regenerate `build/`. Touches no target repo. |
| `evals PROJECT` BILLS | Run the evals of every agent without a current pass |
| `--only ASPECT/AGENT ...` | Limit to these |
| `--max-total USD`, `--jobs N`, `--workdir DIR` | Ceiling, default 30; parallel agents, default 3; workspace, default a temp folder |
| `check PROJECT` FREE | Readiness, and drift against each target's `origin/main`. Exits 1 if not ready or if targets differ. |
| `publish PROJECT` FREE | Open a PR in each target repo that differs. Outward-facing. |
| `--dry-run` | List the changes. Refuses if the project isn't ready. |
| `--trailer TEXT`, `--pr-footer TEXT` | Last lines of each commit message and PR description |

# 16. File formats

Every file is YAML, with a Markdown body for traits, skills and packs. A file's `name` must match its file name or folder. **Bold** fields are required. Three small conventions appear inside values: `{name}` placeholders, `tool.command` references, and list order meaning merge order.

## Agent: `agents/<name>.yaml`

| | |
|---|---|
| **`name`**, `description` | Identity. Deploys use the description for routing, so always write one. |
| `traits`, `skills` | Lists, composed in order |
| **`llms`**, **`preferred_llm`** | Allowed models, and the default, which must be in the list |
| **`prompt`** | The agent's own text |
| **`evals`** | List of `task`, `pass_if`, and optionally `creates`, the new files a task makes. Any `fixtures/...` path in a task must exist unless it's listed under `creates`, in which case it must not exist yet. |
| `requires.capabilities` | What the aspect's tools must provide |
| `hooks` | `{pre_tool_use: [hooks/NAME.py]}` |
| `outputs` | `{field: one-line description}`, read by playbook handoffs |
| `overrides` | `{allow: [TOOL], reason: "..."}` |

## Trait: `traits/<name>.md`

Frontmatter has **`name`**, `tools` and `denies`. A single trait can't both use and deny a tool. The Markdown body is the composed text.

## Skill: `skills/<name>/SKILL.md`

Frontmatter has **`name`**, **`description`** and `requires`, which may hold `tools`, `capabilities` and `min_model_tier` (`haiku`, `sonnet`, `opus` or `fable`). The Markdown body is the composed text.

## Tool: `tools/cli/<name>.yaml` or `tools/mcp/<name>.yaml`

| | |
|---|---|
| **`name`**, **`kind`**, `description` | `kind` is `cli` or `mcp` and must match the folder |
| **`provides`** | At least one capability |
| `commands` | Named shell commands, with `{placeholder}` arguments |
| `server` | For MCP: `command` and `args` |
| `skill` | Path to a skill that teaches the tool, or `null` |
| `artifacts` | Globs of side-effect files to hide from the judge's diff |
| `inspect` | For document fixtures such as `.pen`: `extensions`, `open` and `dump` commands |

## Pack: `packs/<name>.md`

Frontmatter has `name`, `description`, `source` (**`url`**, a full 40-character **`commit`**, `version`, `license`), `layout` (such as `skills: skills/{name}/SKILL.md`) and `base` (`traits`, `skills`).

### Project map: `project-map.yaml`

| | |
|---|---|
| **`project`**, **`catalog`** | Project name, and the catalog path: relative to the map's real location, or absolute. `~` is expanded and symbolic links are followed. |
| `pack` | One pack name |
| `shared_tools` | Tools visible in every aspect |
| **`aspects.<name>`** | **`paths`** (globs), `tools`, `agents`, `preferred_llm`, `pack_exclude` (`traits`, `skills`, **`reason`**) |

### Playbook: `playbooks/<name>.yaml`

| | |
|---|---|
| **`name`**, `description` | Identity |
| **`map`** | Must name the map beside this `playbooks/` folder |
| `inputs` | Names usable as `{placeholder}` in tasks and handoffs |
| **`steps`** | List of **`agent`**, **`aspect`**, **`task`** and `handoff_if`, which every step but the last must have |
| **`evals`** | List of **`inputs`** covering every declared input, **`expect_path`** (the agents you expect to run, a prefix of the steps) and **`pass_if`** |

### Release settings: `release.yaml`

**`target`** is the code root. `claude` maps each destination to a source file. `remove` lists files to delete. `eval_budget` sets a per-run budget by agent.

# 17. Validator rules

The validator runs automatically before evals, playbooks and deploys. These are the checks it makes.

| Rule | What must be true |
|---|---|
| **1** | `preferred_llm` is in the agent's `llms`. An aspect's `preferred_llm` is in every routed agent's `llms`. |
| **2** | A skill's required tools aren't denied by a trait, and the agent's `preferred_llm` meets the skill's `min_model_tier`. |
| **3** | An aspect's tools plus `shared_tools` provide every capability its agents require. |
| **4** | A tool that one trait uses and another denies is listed under `overrides.allow` with a `reason`. |
| **5** | `pack` names exactly one existing pack, and its base can be satisfied by every routed agent. If it can't, the aspect has a `pack_exclude` with a reason. |
| **6** | A playbook names its map; each step's agent is routed in its aspect; each step but the last has one handoff a bound tool command or a declared output can decide; every placeholder is a declared input; and evals exist with a valid `expect_path`. |
| **R1–R3** | `registry-sources.yaml` is well formed; no source has changed since its last sync; imported names don't collide with catalog files or each other. |
| **R4** | A map binds an imported MCP server only if the sources file tags it under `provides`. |
| **R5–R7** | *Warnings:* a bound server's secret variable is unset; a composed imported skill ships files that won't be available; the registry hasn't been synced. |

It also checks that names match file names, no name is used twice, and every referenced trait, skill, hook, fixture, tool and agent exists. Non-executable hooks and overrides that allow nothing denied produce warnings.

# 18. Troubleshooting

## Validator errors

| Message | Fix |
|---|---|
| `agent 'X' requires ['source-control']; bound tools provide ['service-toolchain']` | Bind a tool that provides the capability in that aspect or in `shared_tools`. |
| `aspect preferred_llm 'M' is not in agent 'X' llms` | Remove or change the aspect's model, add the model to the agent's `llms` and rerun its evals, or move the agent to another aspect. |
| `preferred_llm 'M' is not in llms [...]` | Add it to `llms` or choose a listed model. |
| `skill 'S' needs min_model_tier sonnet; preferred_llm 'M' is below it` | Use a stronger `preferred_llm`. |
| `traits conflict on ['Edit']; add overrides.allow with a reason` | Remove the conflicting trait, or add `overrides` with `allow` and `reason`. |
| `name 'X' does not match path (expected 'Y')` | Make `name` and the file name agree. |
| `unknown trait 'X'`, `unknown agent 'X'`, `unknown tool 'X'` | Check the spelling. For an import, run `registry.py sync`. |
| `evals[N] references missing fixture fixtures/...` | Create the fixture, or list the path under `creates:` if the task makes it. |
| `evals[N] says it creates ..., but that fixture already exists` | Delete the fixture or use a new path. |
| `'pack' must name exactly one pack, not a list` | Keep one pack. |
| `agent 'X' is not routed in aspect 'A'` | Add the agent to that aspect's `agents`, after its evals pass. |
| `output handoff on 'F', which agent 'X' does not declare under 'outputs'` | Declare the field in the agent, then rerun its evals, or use a declared field. |
| `expect_path [...] is not a non-empty prefix of the step agents` | List the agents from the first step onward, stopping where the run should stop. |
| `task uses undeclared input(s) ['X']` | Add it to `inputs` or fix the placeholder. |
| `registry is stale for source 'S'; run registry.py sync S` | Run that command. |
| `imported server 'S' provides no capabilities` | Tag it under `provides` in the sources file, then sync. |

## Other problems

| Symptom | Cause and fix |
|---|---|
| `recipe_validate: cannot import the recipe engine` | Set `EPICURA_RECIPE_ENGINE` to the engine's checkout. If the message ends `No module named 'pydantic'`, install pydantic for the interpreter running Kitbash, or set `KITBASH_PYTHON` to one that has it. |
| `refusing to deploy: N agent(s) without a current pass` | Run the `run_evals.py` command printed for each agent, or remove the agent from the map. |
| `no registry at ...; run registry.py sync` | The registry hasn't been built. Run `registry.py sync`. |
| `target is not a directory` | Create the `--target` folder first. |
| `a plugin needs each aspect in one git repo` | An aspect's paths cross repos or aren't in one. Split it into one aspect per repo. |
| An imported MCP server won't connect | Set the variables listed by `registry.py show NAME`. |

| Symptom | Cause and fix |
| --- | --- |
| A released plugin's agents don't appear | Start the session at the repo root, and trust that exact folder. Nothing warns you about this. |
| The judge fails an eval on a huge or irrelevant diff | Run through a map whose tool declares `artifacts`, or add `artifacts` to the tool. |
| A `command` handoff always passes | The tool exits 0 even on failure. Declare a gate command that exits nonzero, as `kicad.erc-gate` does. |
| An eval hits its budget or turn limit | Raise `--budget` or `--max-turns`, or set `eval_budget` for releases. |
| A skill or a tool's command can't be found on your machine | Set the environment variable the tool names, such as `KICAD_CLI`, `ANDROID_HOME`, `JAVA_HOME` or `DEVELOPER_DIR`. `catalog/README.md` lists them all. |
| A recipe eval reports HTTP 429 | YouTube rate-limits caption fetches. Those evals accept an honest failure report. |

# 19. Cheat sheet

Run from `catalog/. py` here means `/usr/bin/python3`. Green is free, amber bills.

### Check

```
py scripts/validate.py
py scripts/validate.py \
    ../projects/NAME/project-map.yaml
```

### Look inside

```
py scripts/run_evals.py AGENT --dry-run
py scripts/run_evals.py AGENT --dry-run \
   --map MAP --aspect ASPECT
py scripts/run_playbook.py PLAYBOOK --dry-run
py scripts/deploy.py MAP --target DIR --dry-run
```

### Evaluate

```
py scripts/run_evals.py AGENT
py scripts/run_evals.py AGENT \
    --map MAP --aspect ASPECT
py scripts/run_evals.py AGENT --budget 0.50 --keep
py scripts/run_playbook.py PLAYBOOK
```

### Create an agent

```
py scripts/make_agent.py "REQUEST" --dry-run
py scripts/make_agent.py "REQUEST" --no-evals
py scripts/make_agent.py "REQUEST" \
    --search "TERMS"
py scripts/make_agent.py "REQUEST" \
   --map MAP --aspect ASPECT --register
```

### Find and import parts

```
py scripts/registry.py sources
py scripts/registry.py search "TERMS" --kind skill
py scripts/registry.py show NAME
py scripts/registry.py sync [SOURCE]
```

### Run a playbook once

```
py scripts/run_playbook.py PLAYBOOK \
    --run KEY=VALUE ...
```

### Deploy

```
py scripts/deploy.py MAP --target DIR
py scripts/deploy.py MAP --target DIR --force
```

### Release

```
py scripts/release.py build PROJECT
```

```
py scripts/release.py evals PROJECT
```

```
# review and merge in the private repo
```

```
py scripts/release.py check PROJECT
py scripts/release.py publish PROJECT --dry-run
py scripts/release.py publish PROJECT
```

```
# merge the PRs, then trust each repo folder
```

## Remember

- **Evals gate registration.** Don't route an agent until its evals pass, and rerun them after changing anything it composes.
- **Changing a map's pack** means rerunning the evals of every agent in that map.
- **After editing `registry-sources.yaml`**, run `registry.py sync`.
- **A clean validator run** ends with 0 errors and 3 warnings.
- **Only `release.py` enforces eval results.** The validator and `deploy.py` don't check them.